

# MAVE

## Multi-Angle Voting for Emission in Closed-Book Knowledge Base Construction

**Jonas Kunze** and Andreas Both

{jonas.kunze, andreas.both}@htwk-leipzig.de

Leipzig University of Applied Sciences (HTWK Leipzig)
Web & Software Engineering (WSE) Research Group
Practice partner: Optimax Energy GmbH

HTWK Leipzig University of Applied Sciences

Web & Software Engineering (WSE) Research Group

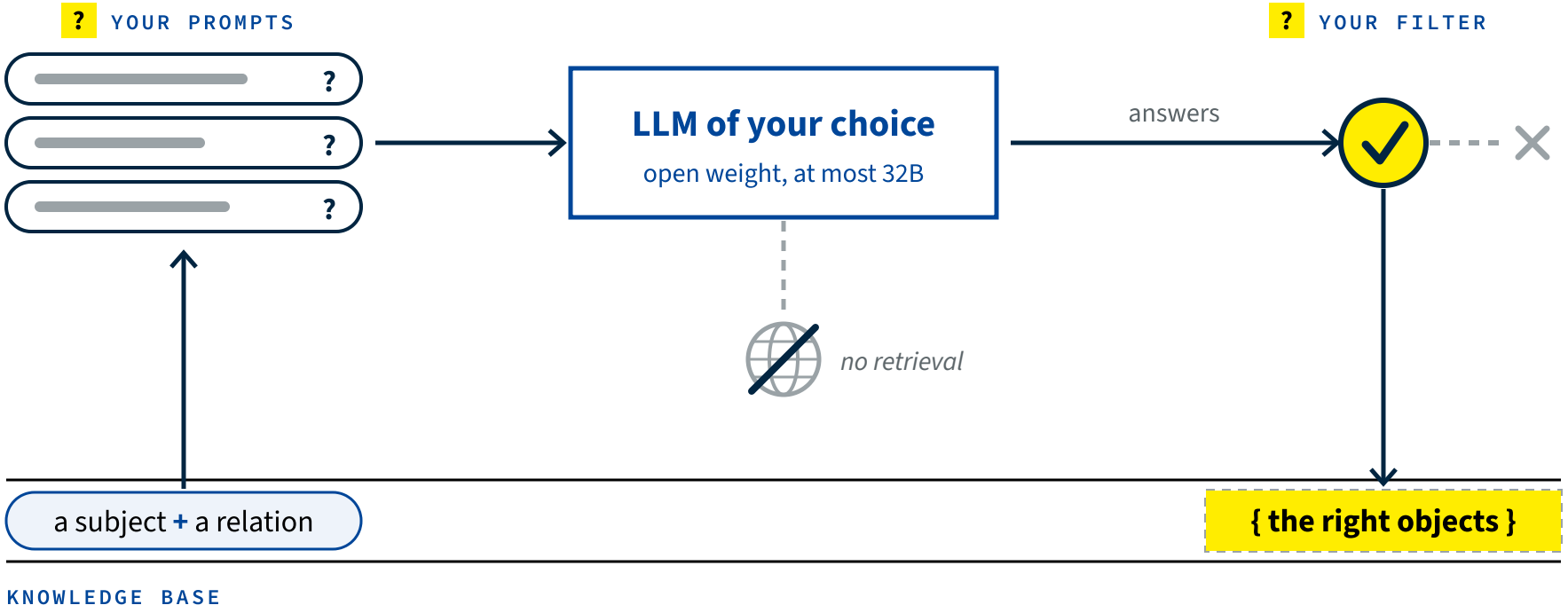
OPTIMAX ENERGY

# The task

Return every object the model memorized, or nothing, for six relations.

SIX RELATIONS

- land borders
- stock exchanges
- award winners
- city of death
- area
- venue capacity





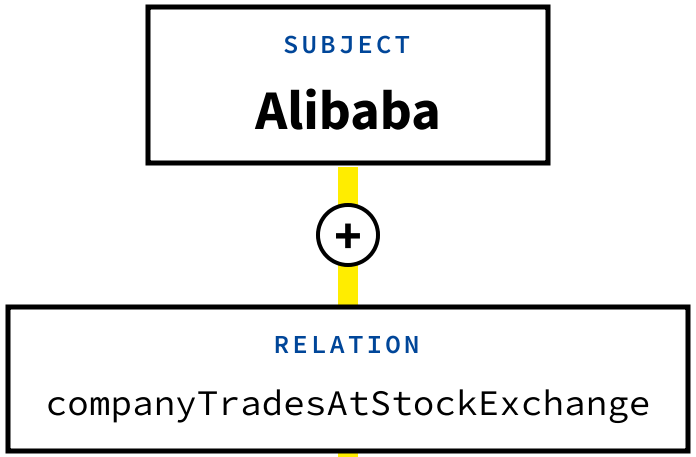
**Jonas Kunze** · jonas.kunze@htwk-leipzig.de
code: github.com/WSE-research/MAVE-LM-KBC-2026

# The task

Return every object the model memorized, or nothing, for six relations.

SIX RELATIONS

- land borders
- **stock exchanges**
- award winners
- city of death
- area
- venue capacity



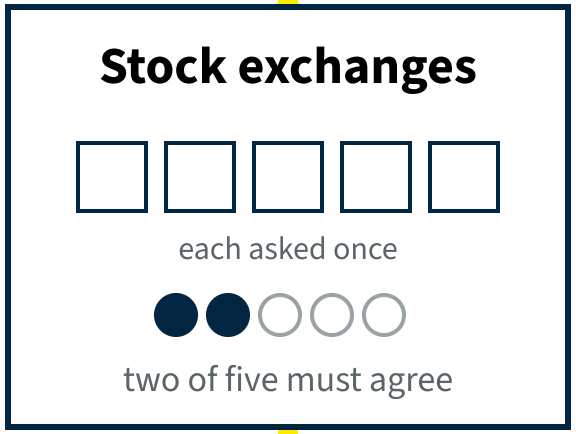

next: the relation picks its recipe



MAVE · LM-KBC 2026 at AKBC, EMNLP 2026
**Jonas Kunze** · jonas.kunze@htwk-leipzig.de
code: github.com/WSE-research/MAVE-LM-KBC-2026

# 1 The relation picks its recipe

which questions to ask,
how often to ask each,
how many must agree




next: ask



MAVE · LM-KBC 2026 at AKBC, EMNLP 2026
**Jonas Kunze** · jonas.kunze@htwk-leipzig.de
code: github.com/WSE-research/MAVE-LM-KBC-2026

# 1 The relation picks its recipe

which questions to ask,
how often to ask each,
how many must agree

## Stock exchanges



each asked once



two of five must agree

## The other five relations

| | | |
|---|---|---|
| Land borders |  |  |
| Award winners |  |  |
| City of death |  |  |
| Area |  |  |
| Venue capacity |  |  |



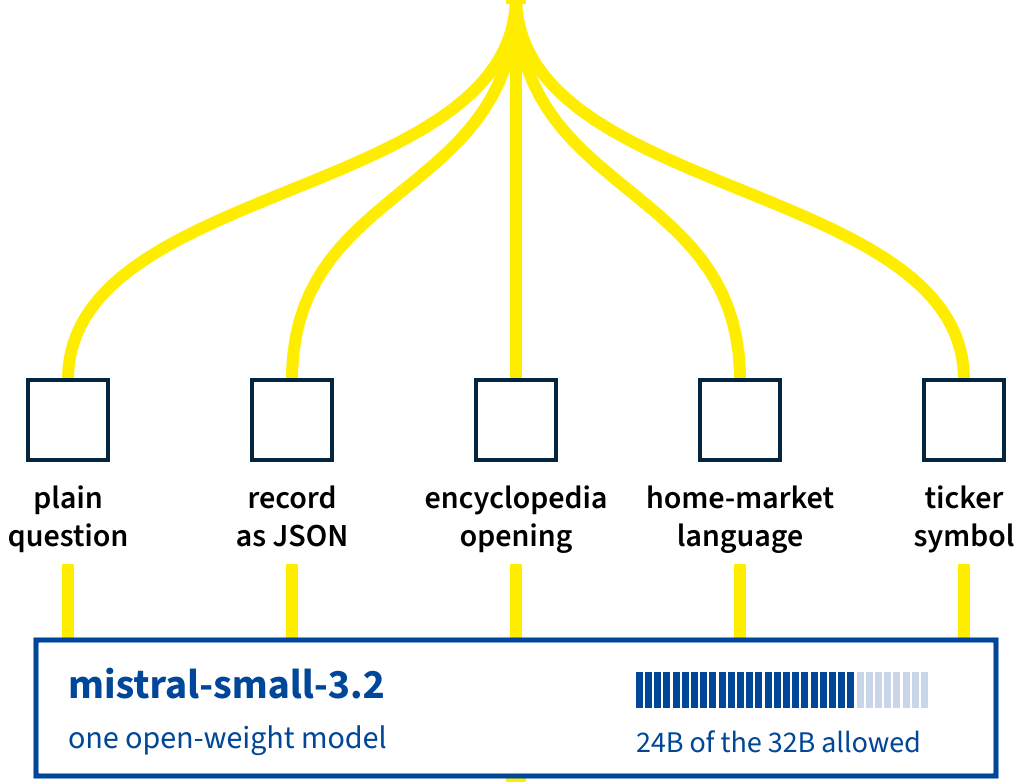

next: ask



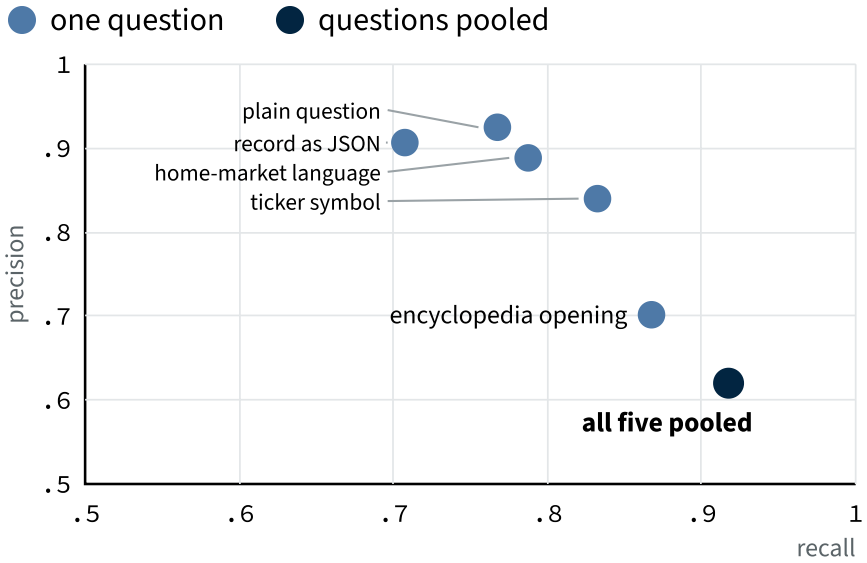
MAVE · LM-KBC 2026 at AKBC, EMNLP 2026
Jonas Kunze · jonas.kunze@htwk-leipzig.de
code: github.com/WSE-research/MAVE-LM-KBC-2026

## 2 Ask

five questions, five kinds of text, all to one model, each once




## Every question finds something else



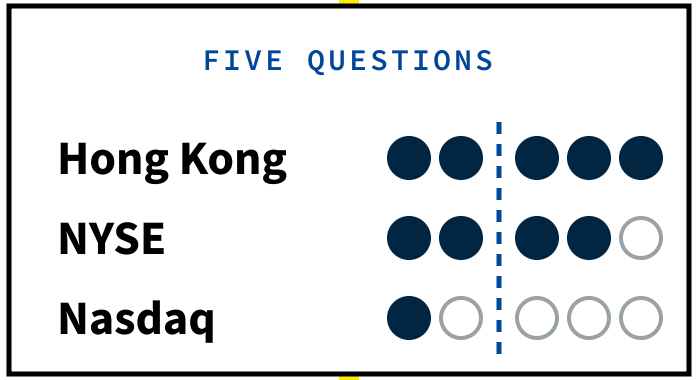

**Pooled, they find more, but more of it is wrong.**

next: pool


**Jonas Kunze** · jonas.kunze@htwk-leipzig.de

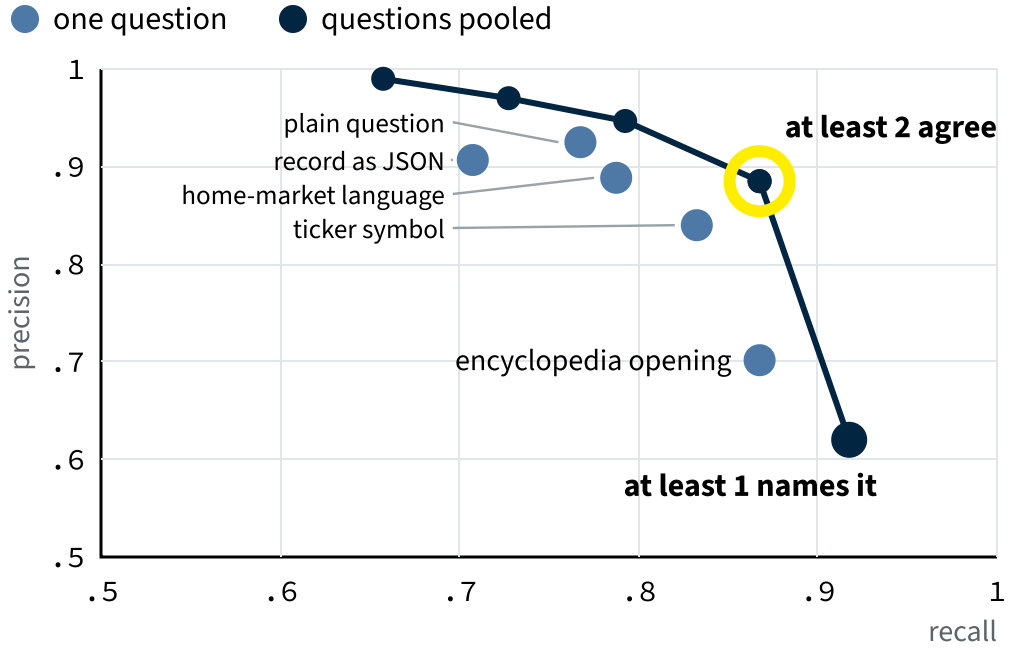
## 3 Pool

each question has one vote, and silence is never a vote against




## Over all companies

validation split, each question alone and pooled




**Asking two to agree buys the precision back.**

next: decide


**Jonas Kunze** · jonas.kunze@htwk-leipzig.de

## 4 Decide and answer

an exchange needs two of the five questions

**2 of 5: passes**

*Nasdaq, named once*

**{ Hong Kong, NYSE }**

# 38%

of the companies are listed nowhere, so there the empty answer is right

validation split

next: score


**Jonas Kunze** · jonas.kunze@htwk-leipzig.de
code: github.com/WSE-research/MAVE-LM-KBC-2026

# 5 Score

macro-F1 over all six relations, hidden test set

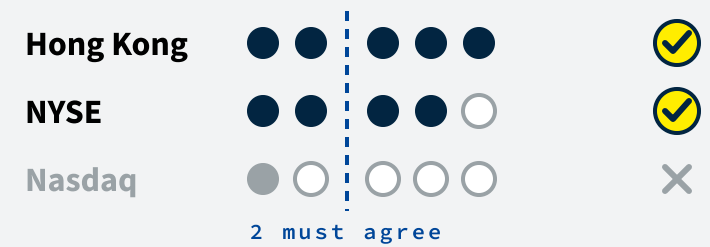
**MAVE 0.6908, second place**

0.3085 baseline: one direct question

0.7060, first place, 31B

## Two things to take home

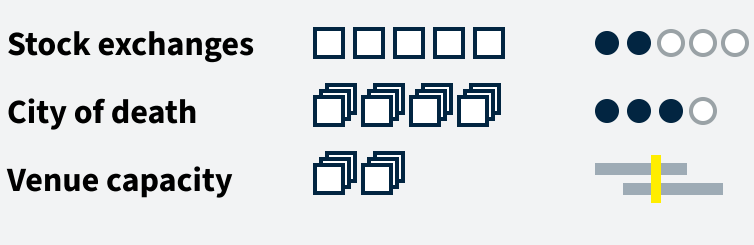
**1**




### Use agreement as a confidence signal.

Pool the answers of several prompts,
then keep the ones enough of them agree on.

**2**




### Tune the recipe per relation.

Each relation needs its own questions
and its own amount of agreement.

**Code, prompts and every run configuration**

`github.com/WSE-research/MAVE-LM-KBC-2026`

Funded by the German Federal Ministry of Research, Technology and Space (BMFTR), grant 03FHP239.


**Jonas Kunze** · jonas.kunze@htwk-leipzig.de

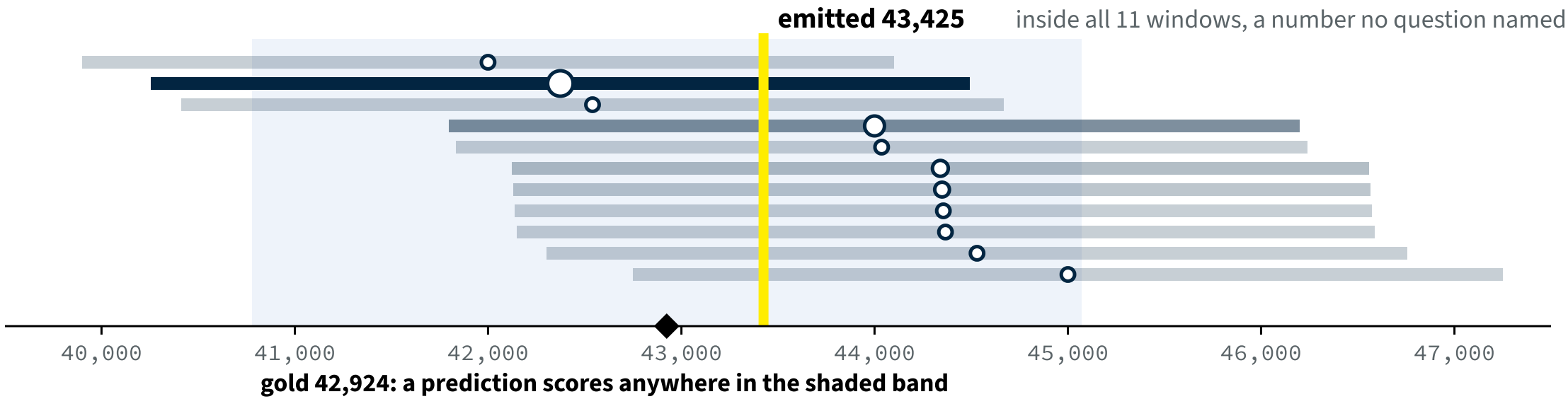
side branch: numbers use the 5% window

{ Hong Kong, NYSE }

next: score

# 4 Decide, for numbers

hasCapacity · Arena Amazônia · train




**42%** of the venues get a hit from a random pool of this size

**68%** of the venues get a hit from the model's own pool

HTWK Leipzig University of Applied Sciences · WSE · OPTIMAX ENERGY


Jonas Kunze · jonas.kunze@htwk-leipzig.de
code: github.com/WSE-research/MAVE-LM-KBC-2026

# B2 When nothing is named twice

IKEA, privately held

**Nasdaq Stockholm** ●○○○○ named by one question

**not listed** ●●●●○ say 4 of the 5 answers

**{ }** the right answer, it is not listed


**Jonas Kunze** · jonas.kunze@htwk-leipzig.de

# B3 F1 per relation and split

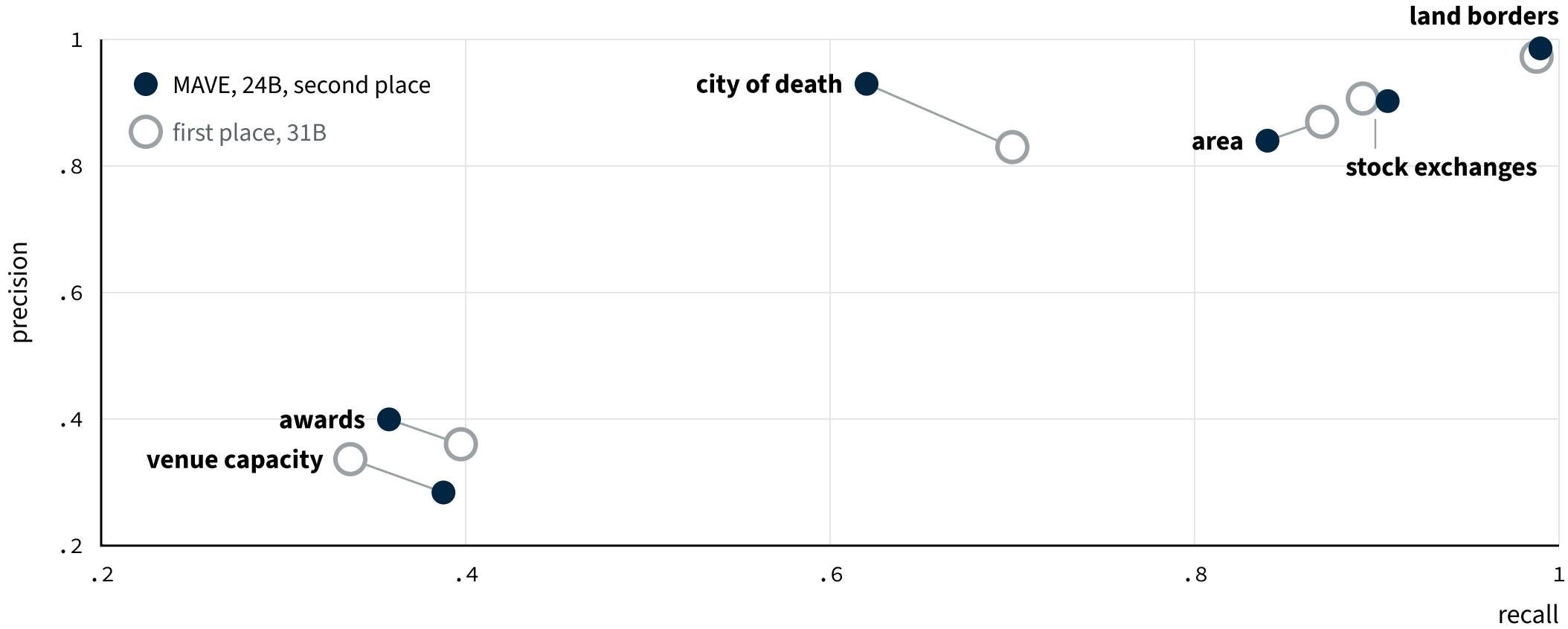
train and validation from our own evaluation, test from the organizers' scorer

| | train | validation | test |
|---|---|---|---|
| Land borders | 0.99 | 0.98 | 0.99 |
| Stock exchanges | 0.73 | 0.78 | 0.85 |
| Area | 0.83 | 0.85 | 0.84 |
| City of death | 0.52 | 0.61 | 0.59 |
| Award winners | 0.51 | 0.24 | 0.36 |
| Venue capacity | 0.25 | 0.26 | 0.31 |
| **Macro-F1** | **0.64** | **0.67** | **0.69** |


**Jonas Kunze** · jonas.kunze@htwk-leipzig.de
code: github.com/WSE-research/MAVE-LM-KBC-2026

# B4 Each relation loses its points somewhere else

official test scores per relation, averaged over items, ours against the first place





Jonas Kunze · jonas.kunze@htwk-leipzig.de
code: github.com/WSE-research/MAVE-LM-KBC-2026

# B5 Land borders: the questions

both lists must name a neighbour · the landmass answer can only remove one

| | |
|---|---|
| **plain question**<br>asked once | Which countries share a land border with *country*? |
| **count and list**<br>asked once | How many sovereign states share a land border with *country*, and which are they? … Count LAND borders only; exclude maritime boundaries. |
| **landmass**<br>asked 3 times, sampled | Which landmass is *country* on, and who else is on it?<br>a country alone on its island has no land neighbours |


**Jonas Kunze** · jonas.kunze@htwk-leipzig.de
code: github.com/WSE-research/MAVE-LM-KBC-2026

# B6 Stock exchanges: the questions

two of the five must name an exchange · if more answers deny a listing than affirm one, nothing is emitted

| | |
|---|---|
| **plain question**<br>asked once | Company: *company*<br>the system prompt asks whether it is listed and on which exchanges, as JSON |
| **record as JSON**<br>asked once | Recall the following facts about the company *company* and return ONLY a JSON object … listing_status, primary_ticker, stock_exchanges |
| **encyclopedia opening**<br>asked once | Write the opening one or two sentences of the English Wikipedia article about the company *company*, … including its stock-exchange listing and ticker … Then add a final line 'LISTED: …'. |
| **home-market language**<br>asked once | the listing question in the language of the home market, e.g.<br>An welchen Börsen ist *company* notiert? |
| **ticker symbol**<br>asked once | What is the stock ticker symbol of the company *company*, and on which exchange does it trade? |


**Jonas Kunze** · jonas.kunze@htwk-leipzig.de
code: github.com/WSE-research/MAVE-LM-KBC-2026

# B7 Award winners: the questions

two of the four must name a person

| | |
|---|---|
| **by decade**<br>once per decade, 1901 to 2029 | Who won the *award* between *start year* and *end year* (inclusive)? List every winner in that period. |
| **open list**<br>asked 8 times, sampled | Who won the *award*? List as many as you can. |
| **by discipline**<br>asked once | Which people working in *discipline* have received the *award*? List as many as you can. |
| **by year**<br>once per year the award existed | Who received the *award* in *year*? List every recipient of that year. |

# B8 City of death: the questions

the city named most is emitted only if more than half of the four questions name it

| | |
|---|---|
| **life sketch**<br>passage 9 times, then read | 1 · Give a short factual account of the life of *person*: … and, if they have died, the date and the CITY of death.<br>2 · the city of death is read from that passage |
| **year first**<br>year 5 times, then asked | 1 · the year of death, as JSON<br>2 · In that year, in which city did this person die? |
| **career**<br>passage 9 times, sampled | 1 · where this person lived and worked over their career, city by city, ending with where they were living when they died<br>2 · In which city did this person die? |
| **final years**<br>passage 9 times, sampled | 1 · Describe the final years and the death of *person*.<br>2 · In which city did this person die? |

# B9 Area: the questions

a five percent window slides over the numbers, and the centre of the window with the most support is emitted

| | |
|---|---|
| **gazetteer entry**<br>passage 5 times, then read | 1 · the feature's gazetteer entry: kind of feature, country and region, coordinates, extent, … and its total area<br>2 · What is the total area of *feature* in km²? |
| **comparison**<br>passage 5 times, then read | 1 · two or three comparable features in the same region, each with its area, then this feature's own area<br>2 · What is the total area of *feature* in km²? |
| **neighbours in area**<br>asked 5 times | Place *feature* in a short table together with the two geographic features that are closest to it in area … End with 'Final answer: <number>'. |


**Jonas Kunze** · jonas.kunze@htwk-leipzig.de
code: github.com/WSE-research/MAVE-LM-KBC-2026

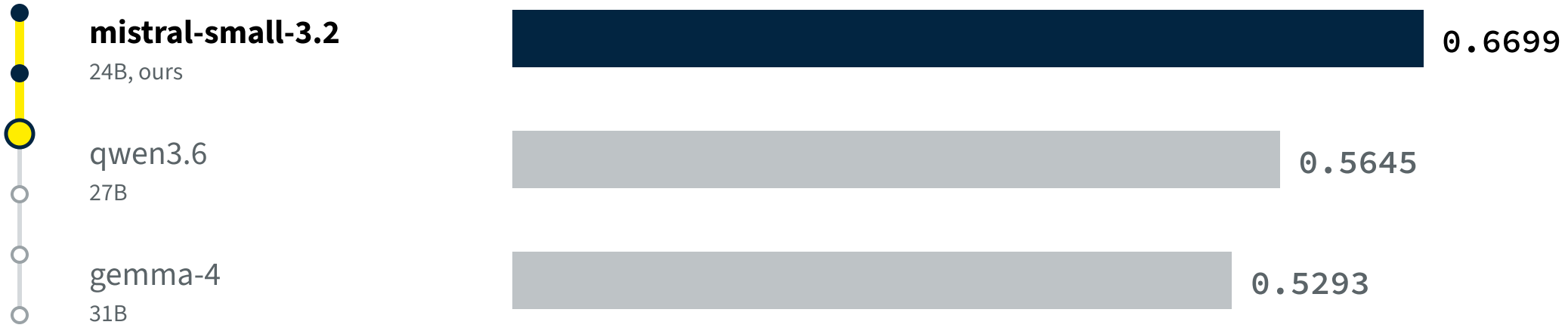
# B10 Venue capacity: the questions

the same window as for area · a second number is emitted when its support is at least half that of the first

| | |
|---|---|
| **encyclopedia opening**<br>asked 20 times, sampled | Write the opening one or two sentences of the English Wikipedia article about *venue* … Then add a final line 'Capacity: <integer>'. |
| **venue record**<br>asked 20 times, sampled | Recall the following facts about *venue* and return ONLY a JSON object … tenant, league, record_attendance, capacity |

# B11 Model comparison

the same pipeline on three open-weight models, every question and rule held fixed




validation macro-F1 · the questions were tuned on the 24B model, which is a bit unfair to the other two

**How the model is asked matters more than its size.**


**Jonas Kunze** · jonas.kunze@htwk-leipzig.de
code: github.com/WSE-research/MAVE-LM-KBC-2026